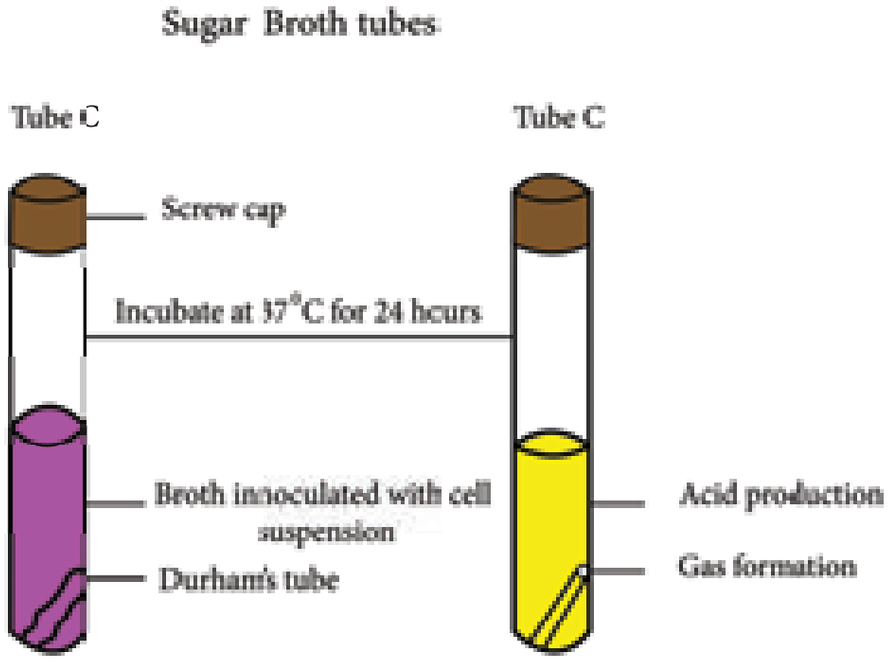
13. Sugar fermentation tube showing acid and gas production

- Carbohydrate broth with bromocresol purple as indicator is used for testing the ability of pure bacterial culture to ferment a specific sugar like lactose, xylose, mannitol and other sugars.
- Acid production is indicated by colour change of the indicator from purple to yellow
- Gas production is indicated by an air bubble in the durham's tube.
- Escherichia coli ferments lactose producing acid and gas.



In the above image Test tube C was innoculated with cell suspension and incubated at 37 °C for 24 hours and after innoculation the purple co-our of broth was changed to yellow colour with gas bubble formation in durham's tube.

14. Agarose gel electrophoresis apparatus.

- Electrophoresis refers to the movement of charged molecules in an electric field.
- The negatively charged molecules move towards the positive electrode while the positively charged molecules migrate towards the negative electrode.
- Gel electrophoresis is a routinely used analytical technique for the separation and purification of specific DNA fragments.
- As the DNA is negatively charged, DNA fragments move through the gel towards the positive electrode. The rate of migration of DNA is dependent on the size and shape.

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